

MULTI-DOMAIN DATA ANALYSIS PORTFOLIO

Executive Summary | Five Complete Python Projects

PROJECTS	5
CLEAN RECORDS	15,986
VISUALIZATIONS	25
AUTOMATED TESTS	15

Portfolio Purpose

This portfolio demonstrates a reproducible workflow from imperfect raw data to validated datasets, descriptive statistics, professional visualizations, responsible interpretation, actionable recommendations, testing, and executive communication.

Cross-Project Overview

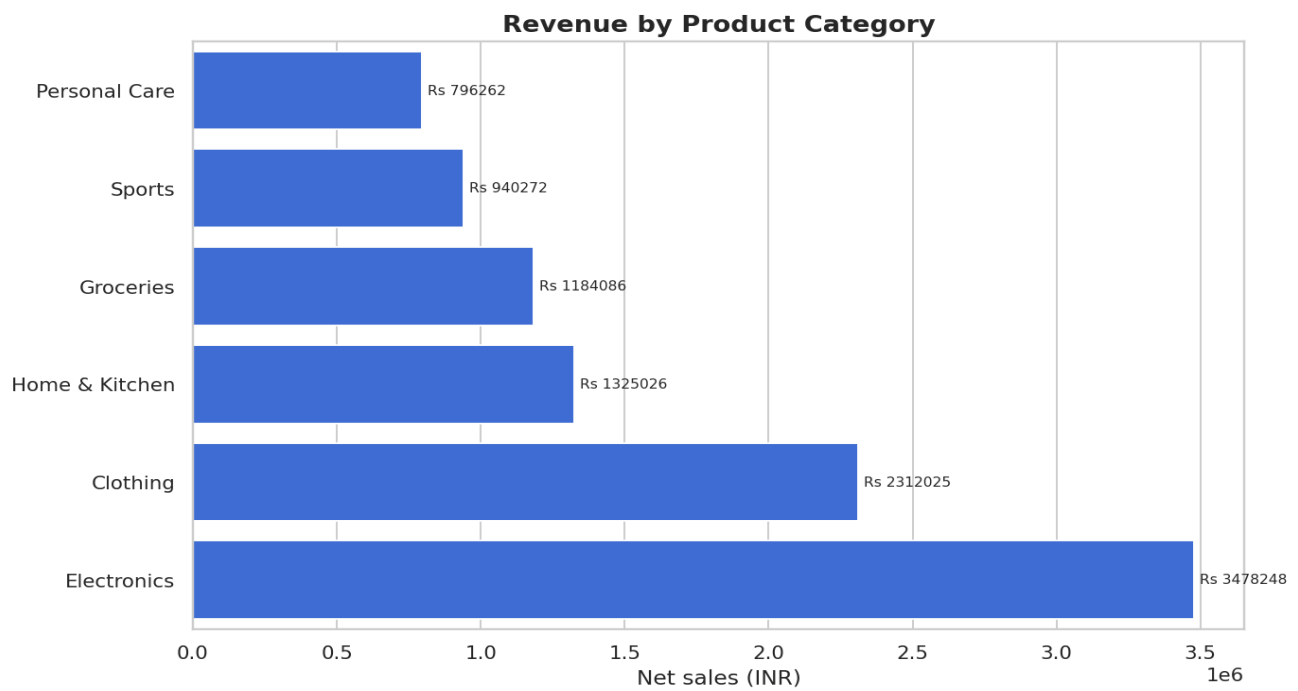
Domain	Clean records	Headline result
Retail	2,500	Rs 10.04M sales
Education	1,200	93.8% pass rate
Weather	6,576	6 cities / 3 years
Healthcare	1,800	50.2% favourable
Finance	3,910	15.1% portfolio return

Shared workflow

Validate schema and quality; clean without overwriting raw data; engineer auditable features; calculate mean, median, correlations, trends, and domain KPIs; visualize evidence; state limitations; test critical logic; generate executive reports.

Retail Analysis

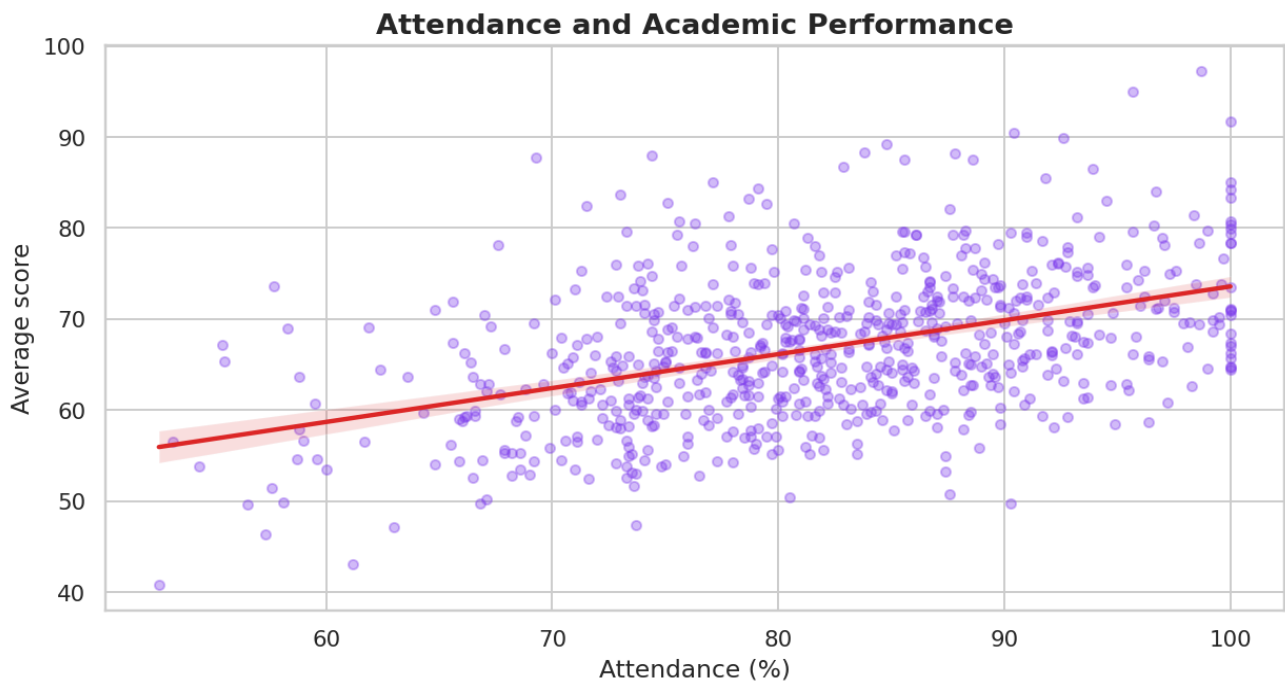
Scale: 2,500 cleaned records **Headline:** Rs 10.04M sales



Interpretation: Electronics leads revenue; Saturday has the strongest average basket.

Education Analysis

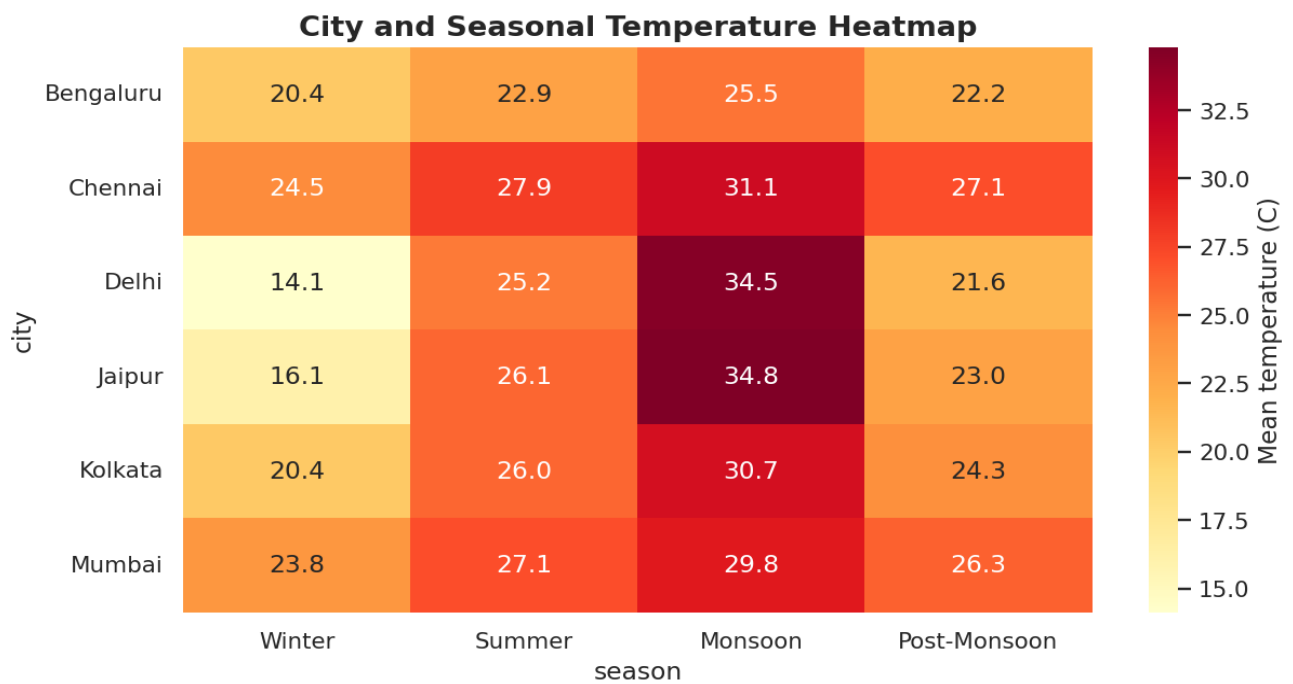
Scale: 1,200 cleaned records **Headline:** 93.8% pass rate



Interpretation: Attendance and study time associate with performance; 321 students are flagged for review.

Weather Analysis

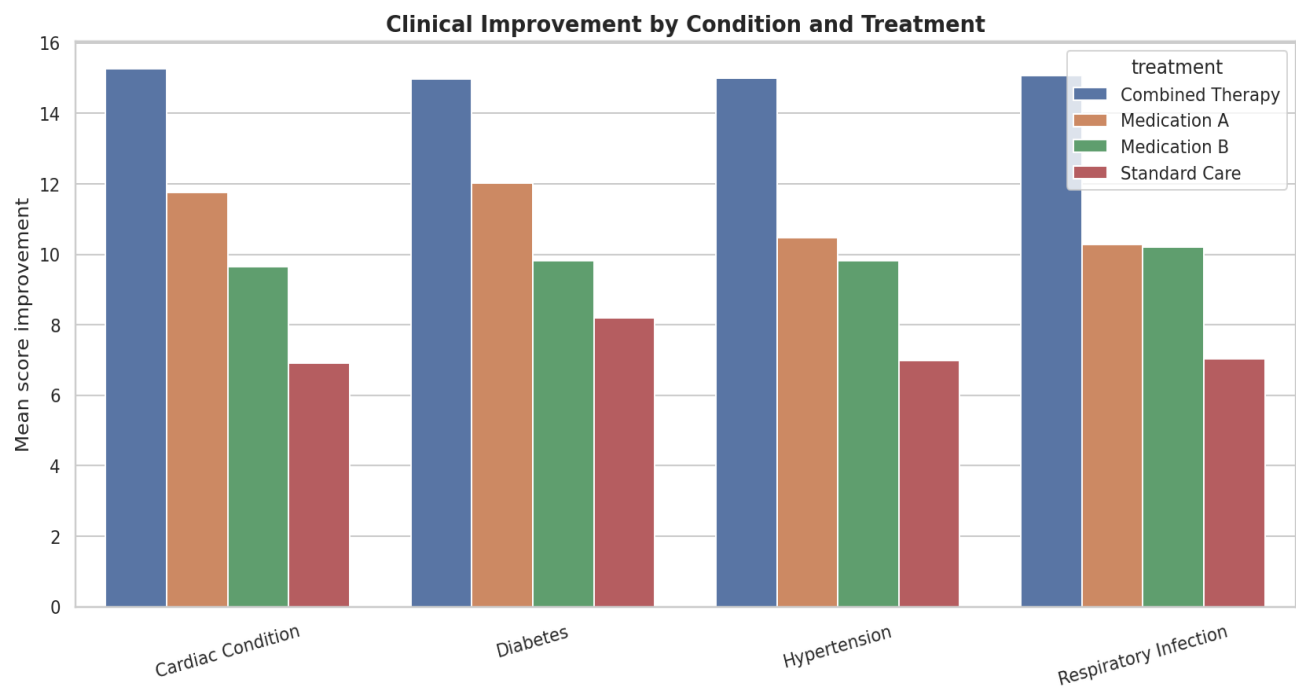
Scale: 6,576 cleaned records **Headline:** 6 cities / 3 years



Interpretation: Chennai is warmest and wettest in the simulation; seasonal exposure differs sharply by city.

Healthcare Analysis

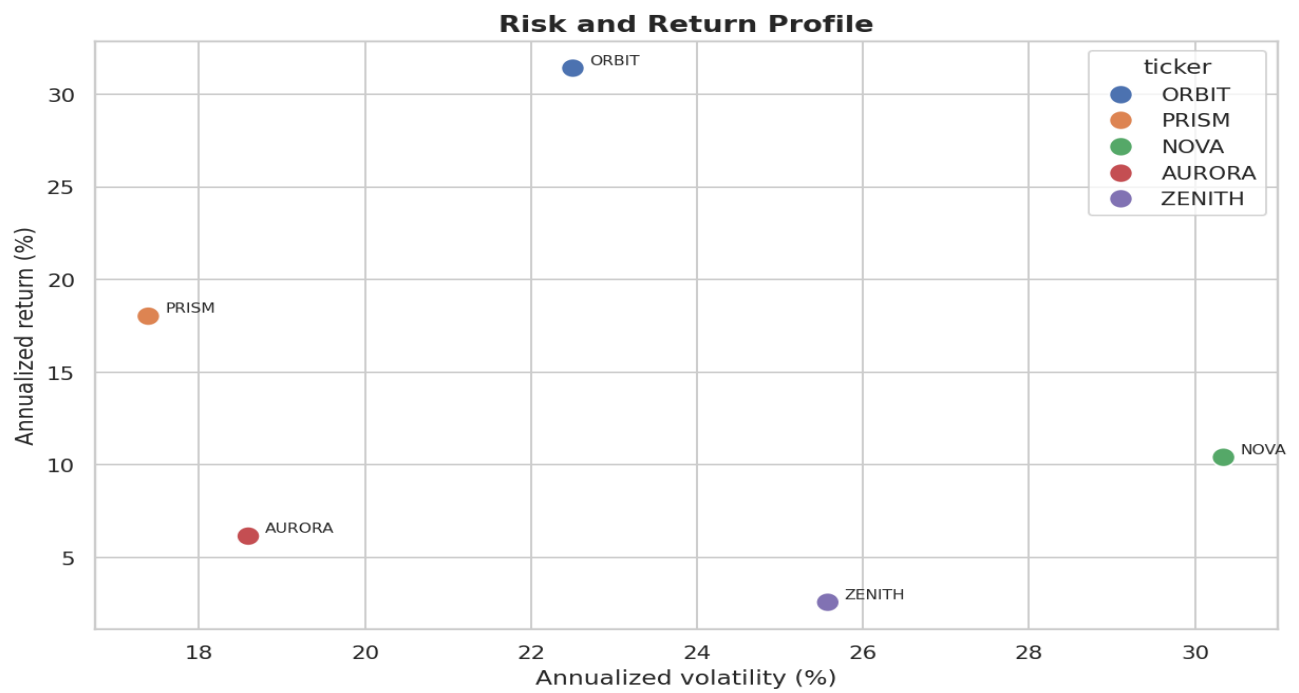
Scale: 1,800 cleaned records **Headline:** 50.2% favourable



Interpretation: Outcome comparisons require condition/severity stratification and remain descriptive.

Finance Analysis

Scale: 3,910 cleaned records **Headline:** 15.1% portfolio return



Interpretation: The simulated equal-weight portfolio shows 15.6% volatility and -18.0% maximum drawdown.

Technical Quality and Responsible Use

Fifteen automated tests reconcile validation, cleaning, feature engineering, classification rules, returns, and KPIs. Fixed seeds make every simulated dataset reproducible, while reusable domain modules separate calculations from presentation.

All data is simulated. Healthcare findings are not clinical guidance; finance findings are not investment advice; weather thresholds are not official warnings; student risk flags require human review. Correlations are associations rather than proof of causality.

Submission Package

Five notebooks, five project reports, twenty-five charts, raw and cleaned data, validation files, reusable Python modules, build scripts, test evidence, complete documentation, presentation deck, demo script, and final checklist.